

Database of Self-Dual Quantum Codes

Introduction

This is a database of objects that have several interpretations:

- Equivalence classes of self-dual additive codes over $\text{GF}(4)$. [1,3,13,14] <#refs>
- Equivalence classes of self-dual codes over the Kleinian four group. [12] <#refs>
- Orbits of simple undirected graphs under local complementation. [1,4,7,8,10,11] <#refs>
- Equivalence classes of isotropic systems. [4] <#refs>
- Equivalence classes of zero-dimensional quantum codes. [5–11] <#refs>
- LU-equivalence classes of quantum graph states. [9,10,11] <#refs>
- Equivalence classes of quadratic Boolean functions under certain transformations. [15] <#refs>

For a more detailed description of the results, see the paper [1] <#refs>, or my Master's thesis [2] <#refs>.

Zero-dimensional $[[n,0,d]]$ *quantum stabilizer codes* corresponds to *additive* $(n,2^n,d)$ codes over $\text{GF}(4)$ which are *self-dual* with respect to the Hermitian trace inner product [3] <#refs>. Zero-dimensional quantum codes represent highly entangled single quantum states.

Every self-dual additive code over $\text{GF}(4)$ can be represented by a *simple undirected graph* [4–11] <#refs>. The graph with adjacency matrix Γ represents the code with generator matrix $\Gamma + \omega I$. *Local complementation*, a simple graph operation, generates the orbits of equivalent self-dual additive codes over $\text{GF}(4)$ [4,7,8,10,11] <#refs>.

Using LC, we can generate the orbits of all inequivalent self-dual additive codes over $\text{GF}(4)$ of length n . These codes have previously been classified by Calderbank et al. [3] <#refs> (up to $n=5$), by Hein et al. [9] <#refs> (up to $n=7$), by H $\diamond$ hn [12] <#refs> (up to $n=7$) and by Glynn et al. [8] <#refs> (up to $n=9$). For higher lengths, only optimal codes have been classified [13,14] <#refs>. We have generated all LC orbits for n up to 12, and this database contains a representative of each orbit. The members of an LC orbit are counted up to isomorphism. (We use the program nauty <<https://web.archive.org/web/20240803110852/http://cs.anu.edu.au/~bdm/nauty/>> to check for graph isomorphism.)

A *decomposable* code can be expressed as the direct sum of two smaller codes. Indecomposable codes correspond to connected graphs. The decomposable codes can easily be constructed by combining indecomposable codes of shorter length. A self-dual additive code over $\text{GF}(4)$ is of *Type II* if all codewords have even weight. We have also classified all Type II $(14,2^{14},6)$ codes.

Tables

Number of indecomposable (i_n) and total number (t_n) of codes:

	1	2	3	4	5	6	7	8	9	10	11	12
i_n	1	1	1	2	4	11	26	101	440	3,132	40,457	1,274,068
t_n	1	2	3	6	11	26	59	182	675	3,990	45,144	1,323,363

These are sequences A090899 <<https://web.archive.org/web/20240803110852/http://www.research.att.com/projects/OEIS?Anum=A090899>> and A094927 <<https://web.archive.org/web/20240803110852/http://www.research.att.com/projects/OEIS?Anum=A094927>> in the On-Line Encyclopedia of Integer Sequences <<https://web.archive.org/web/20240803110852/http://www.research.att.com/~njas/sequences/>>.

Total number of codes of length n and distance d :

$d \backslash n$	1	2	3	4	5	6	7	8	9	10	11	12
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1	1	1	2	3	6	11	26	59	182	675	3,990	45,144
2		1	1	3	4	13	29	107	416	2,618	27,445	615,180
3					1	1	4	11	69	577	11,202	467,519
4						1		5	8	120	2,506	195,456
5											1	63
6												1
All	1	2	3	6	11	26	59	182	675	3,990	45,144	1,323,363

Total number of Type II codes of length n and distance d :

$d \backslash n$	2	4	6	8	10	12	14
2	1	2	5	18	109	2,285	?
4			1	3	19	793	?
6						1	1,020
All	1	2	6	21	128	3,079	$\geq 1,727,942$

See the paper [1] [for more tables and information.](#)

File formats

We make our data available in two formats. Generator matrices for self-dual additive codes over GF(4) are stored in files that can be read by Magma. Graphs are stored in nauty <https://web.archive.org/web/20240803110852/http://cs.anu.edu.au/~bdm/nauty/>'s graph6 format <https://web.archive.org/web/20240803110852/http://cs.anu.edu.au/~bdm/data/formats.html>. This compact representation can be transformed into Magma and Maple format by using the nauty <https://web.archive.org/web/20240803110852/http://cs.anu.edu.au/~bdm/nauty/> package. The graph6 files can be used as input to all the utilities in the nauty <https://web.archive.org/web/20240803110852/http://cs.anu.edu.au/~bdm/nauty/> package. Note that the graph with adjacency matrix I represents the code with generator matrix $I + \omega I$.

Files

One big file

You can download one big file containing all data. The records have the following format:

- length of code
- minimum distance of code
- decomposable or indecomposable code? (Value: D or I)
- type I or type II code? (Value: 1 or 2)
- weight distribution of code
- size of code automorphism group
- LC orbit size
- size of maximum independent set in LC orbit
- graph in nauty's graph6 format

Download

[thebigselfdualfile.txt.gz](#) [thebigselfdualfile.txt.gz](#)

Generator matrices in Magma format

These files contain all codes of length n . For length 14, only extremal Type II codes are included.

n	Download
1	codes1.mag codes1.mag
2	codes2.mag codes2.mag
3	codes3.mag codes3.mag
4	codes4.mag codes4.mag

5	codes5.mag < codes5.mag >
6	codes6.mag < codes6.mag >
7	codes7.mag < codes7.mag >
8	codes8.mag < codes8.mag >
9	codes9.mag < codes9.mag >
10	codes10.mag < codes10.mag >
11	codes11.mag < codes11.mag >
12	codes12.mag.gz < codes12.mag.gz >
14	codes14.mag < codes14.mag >

Graphs in nauty's format

These files contain only connected graphs (corresponding to indecomposable codes). For length 14, only graphs corresponding to indecomposable extremal Type II codes are included.

n	Download
1	vncorbits1.g6 < vncorbits1.g6 >
2	vncorbits2.g6 < vncorbits2.g6 >
3	vncorbits3.g6 < vncorbits3.g6 >
4	vncorbits4.g6 < vncorbits4.g6 >
5	vncorbits5.g6 < vncorbits5.g6 >
6	vncorbits6.g6 < vncorbits6.g6 >
7	vncorbits7.g6 < vncorbits7.g6 >
8	vncorbits8.g6 < vncorbits8.g6 >
9	vncorbits9.g6 < vncorbits9.g6 >
10	vncorbits10.g6 < vncorbits10.g6 >
11	vncorbits11.g6 < vncorbits11.g6 >
12	vncorbits12.g6.gz < vncorbits12.g6.gz >
14	vncorbits14.g6 < vncorbits14.g6 >

References

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